

with lead plates (each weighing more than 8 kg) and other complementary apparatus, like trackers and oscillograms, detects the appearance of tau-neutrinos from the oscillation of muon-neutrinos during their 3-millisecond (ms) travel from Geneva to Gran Sasso.

The OPERA apparatus has been collecting data since 2008, but in March 2011 the researchers noticed something strange. The neutrinos fired from Geneva apparently arrived at Gran Sasso around 60-billionth of a second earlier than light in vacuum!

A tortoise raced the hare

Why should we bother about a speed of this magnitude? Simply because it goes against the founding principle of modern physics that "nothing can travel faster than light in vacuum." If the findings are right, it would wreak havoc on the principle of cause and effect. It would make it possible to send information back in time, blurring the line between past and present. Time travel, space contraction, the conversion of mass into energy and a lot of other things that were once the realm of science fiction would become reality!

The scientists at Gran Sasso conducted more experiments and in September 2011, on achieving similar results repeatedly, they made their findings open. They requested other laboratories to analyse the results and share their opinions, as well as try and replicate the experiments. At the same time, they spotted various areas where their experiment might have failed, with the intention of ruling out all possible weaknesses. In November, they repeated the experiment with improved conditions—in the process ruling out at least one of the potential faults in the first experiment. The neutrino won the race again.

Improved experiment

In the September experiment, the pulses of neutrinos sent by CERN

Tachyons

Tachyons are hypothetical sub-atomic particles that always move faster than light. These were hypothesised by Arnold Sommerfeld and advanced by other scientists. Tachyons have always held the fancy of scientists and science fiction authors. However, till date the concept has been considered only as theory and scientists have held that tachyon particles are too unstable to be treated as existent. Scientists have also held that faster-than-light information transmission and causality violation with tachyons are impossible.

According to Gerald Feinberg, one of the proponents of tachyon theories, even if these are conventional localised particles they would still preserve the basic tenets of causality in special relativity and not allow transmission of information faster than light. So we are yet to see whether the neutrinos observed by OPERA are to be considered as tachyons, and whether they will continue to stick to the theories of special relativity or unveil new truths hitherto unknown!

were relatively long—around 10 ns each. So measuring the exact arrival time of the particles could have large errors. In November, the experiment was repeated with beams that were thousands of times shorter—just 3 ns. There were large gaps of 524 ns between these. So OPERA could measure the arrival time with greater accuracy.

With such short pulses, one does not have to worry about the shape, duration, etc of the beams. It is just a 'click,' and one has to note when it happened without worrying about when the beam started or ended. Being more accurate, a shorter number of readings (20 instead of 15,000 made in September) were enough to replicate the earlier experiment.

While the OPERA team has been making its findings public, it is still reluctant to dispute Einstein's theory. They are merely requesting others to pay attention to this phenomenon and validate the experiments independently, while at the same time planning further experiments themselves. For one thing, they are planning to repeat the experiments using muon detectors with improved clock synchronisation.

The team is also considering the use of optical fibres instead of geographical positioning system (GPS) used currently. Einstein's theory of relativity says that the clock ticks at different rates depending on the gravitational force at a place—this could cause a difference in timing.

The use of optical fibres instead of GPS could overcome this discrepancy also.

Mixed reactions

As with any 'big' announcement, OPERA's revelation has also been met with mixed reactions. While on the one hand, there are scientists raring to create a 'new' physics, there are also those who are dismissing the findings as mere humbug. Faulty experiments, say some, while others blame the different gravitational conditions at CERN and Gran Sasso. Quite understandably, it is not easy for the world of science to accept a challenge to the very foundation of modern physics.

In fact, another group of scientists working at Gran Sasso has itself contradicted the OPERA findings. The ICARUS Collaboration, another team at Gran Sasso, also captured some of CERN's neutrinos. Their goal was to measure the energy and not the journey time. If the neutrinos travelled faster than light, these would have emitted some radiation throughout the journey, losing energy in the bargain. However, the energy remained the same from start to end—so OPERA's claim is not feasible, according to the ICARUS Collaboration.

Prof. Dr Subhash Karbelkar, head of the department of physics, BITS-Pilani, explains, "If an object travels faster than the speed of the appropriate waves, it leaves behind a trail of